

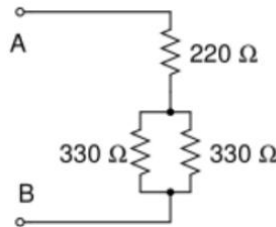
Sample Pre-Lab Quiz Questions for Expt 1

Duration: 10 min, Max marks: 5 marks

1. Out of the following statements regarding Digital Multimeter (DMM), mark **all the correct options**.

- A) The input resistance of a DMM for voltage measurements is typically 10 k Ω .
- ☒ B) For measuring the voltage at node A with respect to node B in a circuit, the **V Ω mA** input terminal of the DMM should be connected to node A and the **COM** terminal to node B.
- C) For measuring the voltage at node A with respect to node B in a circuit, the **V Ω mA** input terminal of the DMM should be connected to node B and the **COM** terminal to node A.
- D) The DMM display showing '1' during a resistance measurement is an indication that the resistance connected between the **V Ω mA** and **COM** terminals is much smaller than the chosen resistance range.
- ☒ E) The DMM display showing '1' during a resistance measurement is an indication that the resistance connected between the **V Ω mA** and **COM** terminals is higher than the chosen resistance range.

2. In the following circuit what is the total resistance in ohms across the terminals A and B?

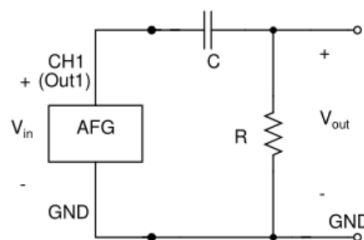


385 Ω

3. With regard to the Digital Storage Oscilloscope (DSO), mark **all the correct options**.

- A) DSOs are versatile electronic instruments used for displaying and measuring time varying current signals.
- ☒ B) DSOs are versatile electronic instruments used for displaying and measuring time varying voltage signals.
- ☒ C) The input resistance of the DSO channel (without using the DSO probe) is 1 M Ω .
- D) The input resistance of the DSO channel (without using the DSO probe) is 1 k Ω .
- E) In order to display a test signal $4 \sin \omega t$ V of frequency 500 Hz on the DSO, the proper scale for the vertical channel is 1 ms/div and horizontal scale 1 V/div.
- ☒ F) In order to display a test signal $4 \sin \omega t$ V of frequency 500 Hz on the DSO, an appropriate scale for the vertical channel is 1 V/div and for the horizontal scale is 1 ms/div.

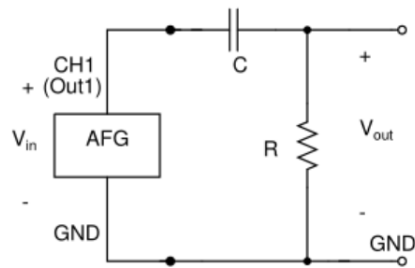
4. The circuit diagram of an RC filter is shown below. Its cut-off frequency was found to be $(1000/\pi)$ Hz. $C = 0.1$ μ F. What is the value of R in ohms?



5000 Ω

5. Circuit diagram of the RC filter circuit of Expt 1 is shown below. Its cut-off frequency is given as f_c Hz. For this circuit mark **all the correct options** from the following.

- A) This circuit is that of a high-pass filter as it attenuates frequencies much higher than its cut-off frequency f_c .
- ✓ B) This circuit is that of a high-pass filter as it attenuates frequencies much lower than its cut-off frequency f_c .
- C) At very low frequencies (i.e. frequencies $\ll f_c$) the gain (ratio of output voltage to input voltage) will be close to unity.
- ✓ D) At very high frequencies (i.e. frequencies $\gg f_c$) the gain (ratio of output voltage to input voltage) will be close to unity.



---- The End ----